

A New Look At General Education

An Excerpt from *Building the Intentional University* (MIT Press, 2017)



MINERVA®

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When we built our first and flagship partner, Minerva University, we wrote a book, *Building the Intentional University* (MIT Press, 2017), detailing exactly how we did it—from curricular philosophy to pedagogical science, and everything in between. It captured the core of our pioneering approach to higher education, and we demonstrated how it can be achieved. While the world has since dramatically shifted, our educational approach was ahead of its time—it was designed to help institutions navigate systemic challenges and prepare learners to thrive post-graduation. We have selected chapter excerpts that we believe will help leaders transform their institutions.

In this chapter, we delve deep into how we redesigned the general education curriculum.

The Minerva Model

What we teach in our general education program is a set of habits of mind and foundational concepts—the HCs¹—that span four core competencies: thinking critically, thinking creatively, communicating effectively, and interacting effectively. This material is structured in a particular way, which constitutes our general education program.

We think of the HCs as constituting not so much a set of things that everyone ought to know but rather a set of tools that everyone ought to use—something akin to a basic cognitive operating system. Our aim is to anchor these tools so robustly in the minds of our students that they reach for them and apply them in all relevant contexts, quickly and naturally. As with the

acquisition of any kind of expertise, this requires a process that begins with conscious deliberation and continues with spaced practice in a variety of contexts, so that is how we teach: we introduce the HCs explicitly (each habit or concept has a hashtag name, e.g., #rightproblem, and accompanying explanatory text; we assign relevant readings and other materials accompanied by study guides; we meet in small, live seminars to practice applying the habit or concept in real-world contexts, to explore its subtleties and address confusions; and we ask students to exercise their nascent skills in independent or team-based assignments.

The principles from the science of learning (see chapter 11)² clearly point away from lecture and

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toward practice using the material, so that is how we organize our sessions; moreover, we ensure that all students are engaged at least 75 percent of the time—and that is what we mean by fully active learning (see chapter 12)². Minerva

seminars always center on thinking things through, making connections, and practicing skills. Only rarely—probably less than 5 percent of the total time spent in class—do they include explicit instructions for information transfer.

TABLE 1
Key Features of Our General Education Program



All first-year students take the same **four courses**, called the “cornerstones” (these are described in detail in chapters 5–8)². Each course lasts the whole freshman year.



Across the cornerstones, there are approximately **115 learning objectives (HCs)** spanning the four core competencies.



Each HC has its own **5-point mastery rubric**. Rubric scores are the primary determinants of overall course grade.



The HCs are illustrated with **content from multiple domains**, driven by Big Questions. (see page 8).



HC **assessment continues in every course** throughout the college career; freshman grades are continuously updated on this basis.

Learning objectives and assessment

The core implementation details for this overarching plan are summarized in table 1. In their first year, all students take the same four “cornerstone” courses, each one introducing approximately thirty HCs. The academic calendar is such that this means that a new habit of mind or foundational concept is introduced in approximately half the class sessions. The remainder of the sessions are either continuation days – needed for particularly rich HCs – or synthesis days, on which bundles of HCs are applied in combination. It is rare for more than one habit or concept to be introduced on a single day, but this does occur occasionally, especially when two HCs are in some sense different sides of the same coin.

We focus on formative rather than summative assessment (i.e., we prioritize low-stakes feedback provided in the early and middle stages of learning over high-stakes

exam scores arriving at the end) and are intensely concerned with transparency and reliability in this process. For this reason, every one of the HCs has its own five-point mastery rubric, each derived from a common template (see table 2). Students receive HC rubric scores on in-class contributions and formal assignments, the latter of which are assigned weights according to their difficulty or importance.

To ensure that students’ commitment to general education goals remains firm throughout their time at Minerva, we have implemented what might be most naturally thought of as “time-traveling” grades. In sophomore and later years, students receive grades for the learning outcomes related to course content and grades related to their use of relevant HCs. HC grades from later years are more heavily weighted than those from the first year, so students are incentivized to demonstrate and improve their mastery continually. At graduation, the letter grade for each cornerstone course – a function of the accumulated grades for all the HCs it introduced – is finally locked in, representing each student’s ability in each of the four core competencies as evinced over their entire college career.

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Throughout this process, students are frequently reminded of their mastery of the general education goals. The Active Learning Forum's dashboard (see chapters 15 and 16)² displays each student's changing scores across the entire set of HCs. These can easily be sorted according to either course of introduction or core competence. A summary report at the core-competence level also appears on term reports, and we have plans to include it as an option on per-student customizable employer-facing CVs.

TABLE 2
Mastery Rubric Template

Rubric score	Description
1 Lacks knowledge	Does not recall or use the learning outcome when prompted or does so mostly or entirely inaccurately.
2 Superficial knowledge	Recalls or uses the learning outcome only somewhat accurately, by partially quoting, paraphrasing, summarizing, outlining or applying it, or recalls or uses the skill or concept in ways that fail to address the relevant problems or goals.
3 Knowledge	Accurately recalls, uses, paraphrases, summarizes, outlines, or reproduces standard or straightforward examples of the learning outcome and does so in a way that addresses the relevant problems or goals.
4 Deep knowledge	Demonstrates a deeper grasp of the learning outcome by explaining it, using it to produce a sophisticated, nonstandard example, differentiating component parts, or applying critical distinctions, or by analyzing relationships between component parts.
5 Profound knowledge	Uses the learning outcome in a creative way, relying on a novel perspective (i.e., not one that was in course materials or is easily located in the relevant literature) to: improve an existing problem-solving technique or create a more effective one; devise a more elegant or beautiful solution than the standard; or produce an unusually clever and effective application.

Using Big Questions to organize content

We realized early that even if one wishes to focus on transferable skills and knowledge rather than overt knowledge of specific content, one still must establish some kind of organizational scheme for the content. There are several good ways of doing this, but earlier work (Fost, 2013) suggested a natural solution: use so-called Big Questions. The idea came from a recommendation by the Association of American Colleges and Universities (2007) that curricula should “engage the big questions” by “[teaching] through the curriculum to far-reaching issues, contemporary and enduring, in science and society, cultures and values, global interdependence, the changing economy, and human dignity and freedom.”

The advantages of this approach are clear. First, by associating questions with instructional units – that is, natural groupings of HCs – that last two to three weeks, we ensure that the subject matter for a certain course will remain relatively stable. This strategy decreases the time and effort required to learn content related to a narrow domain and keep it in mind long enough to participate in classwork, only to discard it and repeat the process with another subject. Minerva students spend correspondingly more effort focusing on the HCs rather than on frequently changing content incidental to the learning objectives.

Second, the questions provide real-world domains of application. The first of these is the Big Question associated with the unit in which the habit of mind or foundational concept is introduced. Later in the course, however, synthesis lessons bundle several HCs and apply them anew to a different question. This can occur multiple times within a course, or even between courses. For example, the Big Questions “How can we feed the world?” and “Can war be avoided?,” among others, appear in several cornerstone courses. This approach harmonizes with several principles from the science of learning (see chapter 11)², including spaced practice, interleaving, and building on prior associations.

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Command central:
The curriculum map

To see the cornerstone courses under even greater magnification, let’s examine a small piece of what we call the curriculum map. This map shows, for all four freshman year long courses, which learning objectives will be addressed and when, using content from a particular Big Question. The excerpt shown (see table 3) is from the first two weeks, fall semester, of the Empirical Analyses course.

Each row in the “HC(s) introduced” column in table 3 represents a single class session. Thus the first session of the Problem Solving unit introduces the habit of mind #selflearning, which is about techniques to facilitate self-study and unsupervised learning. In the next two sessions, four more HCs are introduced. Readings for all five

relate either to the HCs themselves or to some aspect of the Big Question that will permit illustrations of its application. For example, in session 3 of this unit, students might read an article about how a nonprofit group approached providing agricultural assistance to an underdeveloped country. The study guide for that reading might ask students to focus on how the group used gap analysis (the focus of the #gapanalysis foundational concept) to determine which elements of the problem could be addressed with existing tools and which elements required new solutions. In class, students might share their understanding of this process, then work in groups to apply gap analysis to a slightly new problem, still related to the same Big Question, “How can we feed the world?”

TABLE 3
Excerpt from the Curriculum Map Showing the First Two Weeks of Empirical Analyses Course

Big Question	Unit	HC(s) introduced
How can we feed the world?	Problem Solving	#selflearning #rightproblem; #breakitdown #gapanalysis; #constraints <i>Synthesis: #rightproblem, #breakitdown, #gapanalysis, #constraints</i>

How We Met Common Challenges

In the scholarly fora of general education – places like the pages of The Journal of General Education – discussions of curriculum structure are often accompanied by others equally important: those concerning implementation. In this section

I review some of the ways that we either avoided or overcame structural and sociopolitical challenges faced elsewhere. As maps of the terrain, I will use Zai (2015) and Newton (2000).

OVERCOMING COMMON CHALLENGES

Challenge	How We Overcame
Lack of rigor/ watered-down courses	- All students take the same courses in the first year and are assessed on the HCs throughout their undergraduate career. - Big Questions function as cross-course bonds, explicit demonstrations that a single real-world problem typically requires the synthesis of ideas from different core competencies.
Troubled leadership and reform	- Faculty performance is measured first and foremost by professors' successful teaching, rather than research. All faculty teach a cornerstone course at least once and receive more than a month of training, which includes the science of learning and pedagogical best practices.
Who can or should teach	A team approach is adopted where all faculty teaching interdisciplinary courses receive lesson plans and are accompanied by a subject matter specialist prior to teaching. - Standardized lesson plans are written by a core team of interdisciplinary faculty.
Multidisciplinary or transdisciplinary knowledge	We deemphasize, in our general education curriculum, knowledge acquisition and instead favor teaching habits of mind and foundational concepts that either help produce knowledge or help students deploy knowledge in the service of a goal.

01**Lack of rigor/watered-down courses**

One set of challenges concerns academic standards and rigor. Sometimes these challenges have to do with the coherence and conceptual unity of the curriculum itself, or the degree to which curricular rigor is enforced by an intellectual community that shares a “language of ideas” and a commitment to policing the integrity of those ideas. Other concerns center on delivery: general education classes are commonly taught in large classrooms and by the least expensive (and often least experienced) faculty.

Regarding coherence, we focus on two specific issues. The first is interpersonal, namely, the extent to which one can say something definite about the recipient of a degree. One of the historical purposes of general education

programs was to establish a standard level of competence and knowledge that every graduate would achieve. Prospective employers are interested in that, but so are current and prospective students. In this sense, the Minerva program coheres because everyone takes the same courses in the first year and continues to be assessed on the HCs throughout their undergraduate career. The cornerstone course grades that appear on a student’s transcript guarantee that external audiences, and the students themselves, can compare student ability in the four high-level core competencies.

Another key issue regarding coherence is cross-course synthesis, or the degree to which the curriculum itself hangs together. This concern has a history dating

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back to about 1820 (Boning, 2007), which was when classical liberal arts programs began to meet competition from programs (such as those at Bowdoin, Brown, or Michigan) that offered more individualized pathways to help students prepare for specific vocations. The Minerva approach to this issue is multifaceted. As mentioned above, some Big Questions appear in multiple courses. “Can war be avoided?” is one example. These questions function as cross-course bonds,

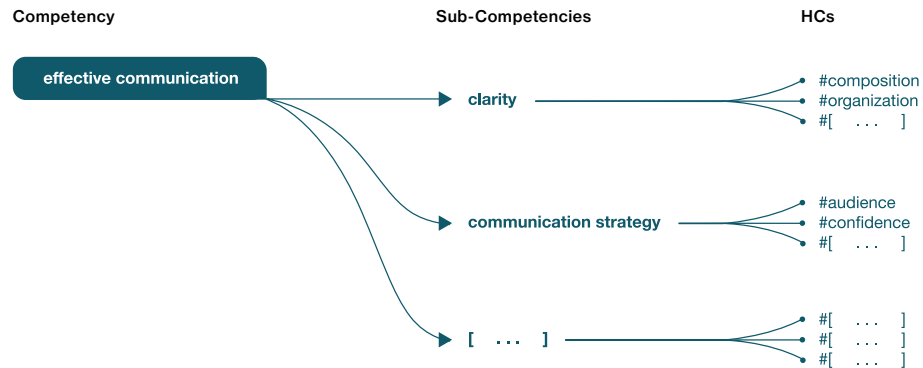
cornerstone (eight total), combining them into a coherent approach to their chosen topic. Students complete two of these projects, one in fall semester and another in spring semester, and this work is the most heavily weighted when we calculate the students’ HC grades. Last, as students progress through the year, they employ HCs introduced in one cornerstone course across the other cornerstone courses on a regular basis. In fact, there are sections in some cornerstone courses that cannot be taught – at least not optimally – without students having taken segments of other cornerstone courses.

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explicit demonstrations that a single real-world problem typically requires the synthesis of ideas from different core competencies. Another way we achieve the same end is through our use of final project assignments. Final projects consume the last week of each cornerstone course and require students to invoke at least two HCs from each

The remainder of the academic standards concerns have already been addressed elsewhere in this chapter, if implicitly. Conceptual unity arises from how we have organized the HCs into core competencies (see chapter 2)². There is certainly an intellectual community for the HCs, in the sense of a group of people who share a language of ideas and attend assiduously to one another’s use of them. One of the requirements for a habit or concept is that there be a scholarly literature about it, but in many ways we are creating a new community for whom this structure will become a touchstone. The appellation process, that is, the granting of a hashtag name for a habit or a

EXAMPLE Hashtag Names in a Skills Taxonomy



concept, provides that community with a common language. Indeed, even after our first year of teaching, and more so now after three years of student enrollment, it is common to hear HCs dropped by name into casual conversation.

The integrity of the proper application of these concepts and habits is facilitated by scoring rubrics, which detail exactly what we say it means to apply a concept poorly or well, and our explicit definitions, which are displayed prominently on the Active Learning Forum. There is very little room for hand waving about any habit or concept; indeed, our sharpest students have helped draw attention to vagaries in some of our early definitions, prompting revision or correction.

02 Troubled leadership and reform

Zai reports, with concern, that when programmatic reform is led by tenured faculty, these individuals tend to be more committed to their disciplines, or even to communities outside the university (such as professional societies), than to general education. They also tend to be unfamiliar with the relevant literature. At Minerva, we have intentionally avoided the incentive structures that at many institutions push faculty toward research and away from teaching. In the first place, we don't use publication quotas or impact factors to determine professorial retention or advancement. Faculty research is encouraged and

Faculty performance is measured first and foremost by professors' successful discharge of instructional duties—this is why we say that we are student-outcome-centric.

supported, especially to the degree that it can include students as authentic contributors, but it is not the centerpiece of professorial success. Research is also so heavily incentivized within the broader profession that additional pressure imposed by the institution seems counterproductive. Second, and related, faculty performance is measured first and foremost by professors' successful discharge of instructional duties – this is why we say that we are student-outcome-centric. The general education courses are treated no differently in this regard from the disciplinary courses within the colleges. Finally, all our faculty currently teach a cornerstone course at least once and receive more than a month of training, which includes extensive sections on the science of learning, pedagogical best practices, and our general education curriculum.

Zai also reports that lack of academic rigor in some programs may have an organizational cause, namely, the low esteem general education holds in the eyes of the university's power elite. This is not a concern at Minerva. The primary architects of our program are the founder and the chief academic officer, who developed these plans in full view of the board of directors and the funding bodies. It was not merely that these entities approved of the program incidentally, it was that the program was a central reason for their endorsing the overall academic mission.

Another advantage we have with respect to reform is that we are developing our curriculum sequentially, from the freshman year forward. Because the freshman year is entirely focused on general education, to prove we were delivering on our promises we had to do our best work on those courses. There simply was no other university activity to showcase.

03

Who can or should teach

Each of the cornerstones is interdisciplinary, so it is rare for any individual instructor to be thoroughly familiar with all the material in it – either the HCs or the Big Questions. To support them, we hold weekly hour-long meetings with all instructors who are teaching a given course at that time. Prior to the meeting the instructors review the lesson plans to be used two weeks hence, and in the hour allotted they work through any unfamiliar material or pedagogical concerns. Although the lesson plans are already complete and highly polished by the time the instructional faculty receive them, they are flexible enough to accommodate new suggestions and approaches. Also attending the meeting is a specialist – the same person for all four courses – whose role is to help each teaching team understand how its class relates to the other three. That person also passes along institutional memories as needed, including common confusions, best practices for certain topics, and the like. In some cases the dean of the college also attends these meetings.

The team-based approach is also used to develop the lesson plans in the first place. That development was managed by experienced interdisciplinary educators and the lesson plans were heavily edited. Altogether, approximately 3,400

pages of curriculum were written and reviewed by a team of eight to fifteen people over the course of two years. By using such a cross-disciplinary team to develop standardized lesson plans, we further mitigated the impact of any given professor's particular training. Nevertheless, the lesson plans are written in a way that allows us to capitalize on that training: a portion of every lesson plan is left completely free so that the class can explore particularly fruitful student questions or issues that many of them found confusing. In these moments, some professors' expertise will come to the fore. Critically, we use the weekly meetings to capture their handling of the situation and add it as an annotation to the lesson plan so that future students and instructors can benefit from it.

04

Multidisciplinary or transdisciplinary knowledge

On the matter of whether general education ought to include knowledge within multiple subjects or knowledge that transcends subjects, our choice has by now become clear: we deemphasize, in our general education curriculum, knowledge acquisition – that is, knowledge in the sense of a codified set of justified, true beliefs – and instead favor teaching habits of mind and foundational concepts

(see chapter 2)² that either help produce knowledge or help students deploy knowledge in the service of a goal. Consider, for example, the nine-week-long Scientific Method unit in Empirical Analyses or the Leadership and Teamwork unit (also nine weeks long) in Complex Systems. The learning objectives in these units include, respectively, HCs such as #plausibility, which is about the process of determining the internal and external consistency of a hypothesis, and #powerdynamics, which is about types of power and different ways and consequences of wielding it. We assign readings from the scholarly literature on these topics, but our goal is not to ensure that students remember the content of these readings. Even if that were our purpose, the brutal fact is that they would not remember much of it, as the evidence of multiple studies on higher education show. Instead, our purpose is to instill a habit of

mind – that is, a regular practice that becomes automatic in the relevant contexts – or to teach a foundational concept that helps make sense of a problem. This aim is supported by the rubrics, which move quickly past mere demonstration of knowledge and toward competent praxis. A student cannot achieve mastery or receive an A grade by regurgitating information in an exam booklet: we have no exams in these courses. Rather, the student must deploy the skill or concept in the contexts it is needed, which are likely to be classes that have nothing whatever to do with their original context. Those classes, and those contexts, may occur years after the #plausibility habit was introduced. Thanks to our time-traveling grades and Active Learning Forum dashboard, the enduring importance of these general education goals remains in focus throughout students' college careers.

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Conclusion

I conclude with a view expressed by Clifford (1877), written at a time – perhaps not coincidentally – when the diversity of views about general education was in its first flush.



The student who begins to learn about electricity is not asked to believe in Ohm's law: he is made to understand the question, he is placed before the apparatus, and he is taught to verify it. He learns to do things, not to think he knows things; to use instruments and to ask questions, not to accept a traditional statement **If Ohm's law were suddenly lost and forgotten by all men, while the question and the method of solution remained, the result could be rediscovered in an hour.** But the result by itself,

if known to a people who could not comprehend the value of the question or the means of solving it, would be like a watch in the hands of a savage who could not wind it up In regard, then, to the sacred tradition of humanity, we learn that it consists, not in propositions or statements which are to be accepted and believed on the authority of the tradition, but in questions rightly asked, in conceptions which enable us to ask further questions, and in methods of answering questions.

Endnotes

¹Habits of Mind and Foundational Concepts is a Minerva term that designates the cognitive skills that with practice come to be triggered automatically ("habits of mind"); and fundamental knowledge that is broadly applicable ("foundational concepts"). In depth definition and analysis of these concepts can be found in chapter 2 of the book, *Building the Intentional University* (MIT Press, 2017).

²Refers to the full original publication, *Building the Intentional University* (MIT Press, 2017).

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On the Cover: A vortex of books represent a shift from traditional knowledge silos to modern wisdom's light.

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